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H&M Customer Intelligence

Phase 1 Report

Before the trends, the segments and the predictions, there is the data. Phase 1 gets 31.8 million purchases ready to talk.

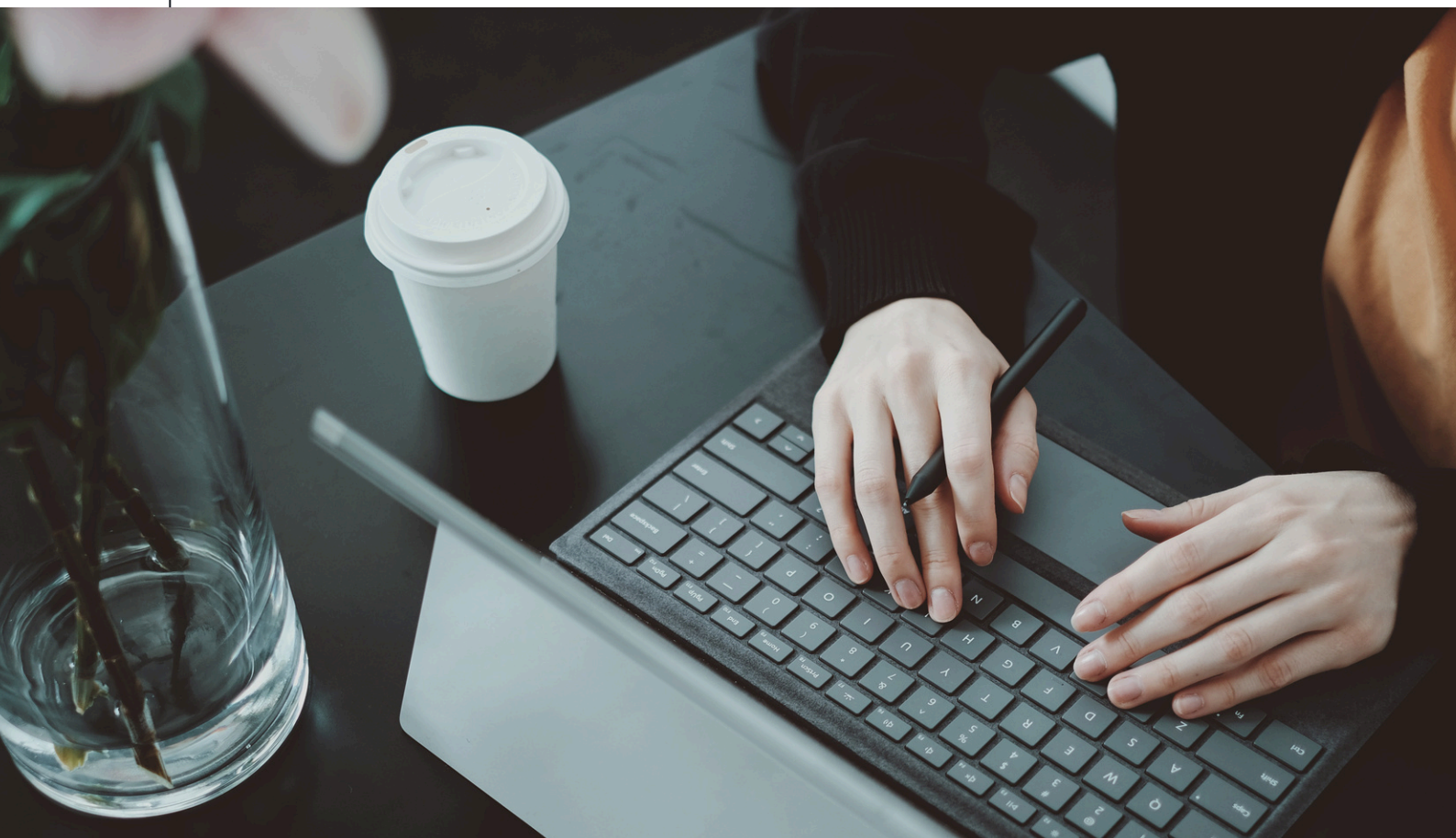
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Executive Summary

Phase 1 tested the quality of all three H&M tables: 1.37 million customers, 105,542 products and 31.8 million purchases. There are no duplicate customers or products, no missing values in purchases, and every purchase links to a known customer and product.

One thing to watch: 9.4% of purchase rows repeat exactly, and were kept as multiple units. With the data now trustworthy, Phase 2 asks what sells, who buys and who comes back.



Main outcomes :

- Dropped FN: 99.8% same info as fashion news frequency
- Articles trimmed from 25 to 15 columns (10 repeated ID columns removed)
- No broken links between purchases, customers and products
- 9.4% repeat purchase rows kept as multiple units

Data Quality :

Table	Rows	Cols before	Cols after	Saved As
customers	1,371,980	7	6	customers_clean.parquet
articles	105,542	25	15	articles_clean.parquet
transactions	31,788,324	5	5	transactions.parquet
customer baskets (derived)	one row per customer per day	n/a	n/a	customer_baskets.parquet



Conclusion & Closing

Tech -Stack

- *DuckDB + SQL*: querying 31.8M rows directly
- *Parquet*: compact, fast storage
- *Python (Colab)*: running and organising the work.

Assumptions

- Prices are scaled, not real currency
- Repeat rows treated as multiple units
- Sales channels are unlabelled



Contact

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